

■ Build a Simple Machine Learning Model

with scikit-learn

Grade 6 · AI & Digital Subject

Learn how computers learn from data and make smart predictions!

■ Lesson Objectives

- Understand what Artificial Intelligence and Machine Learning are
- Learn how ML models learn from data
- Explore the scikit-learn Python library
- Build a fruit-classifier model using a Decision Tree
- Interpret model accuracy and predictions

■ Section 1: What is Artificial Intelligence?

Artificial Intelligence (AI) means teaching computers to **think and learn like humans**. AI allows machines to perform tasks that normally require human intelligence — such as recognising speech, understanding images, and making decisions.

■ AI in Daily Life

■ Voice Assistants

Alexa and Google Assistant understand what you say and reply intelligently.

■ Smart Games

Games that observe how you play and adapt their difficulty to challenge you.

■ Face Recognition

Apps and phones that recognise who you are by analysing your face.

■ Section 2: What is Machine Learning?

Machine Learning (ML) is a **branch of AI** that helps computers learn from data without being explicitly programmed for every task. Instead of writing rules by hand, we show the computer many examples and it figures out the patterns on its own.

■ *Think of it like school: you learn from many examples (practice problems, reading, observation) until you understand the pattern well enough to answer new questions!*

■ Why is Machine Learning Important?

■ Video Recommendations

ML suggests videos and songs based on what you enjoy watching and listening to.

■ Healthcare

Doctors use ML to detect diseases early and help patients get better faster.

■ Self-Driving Cars

ML helps cars see the road and make driving decisions without a human driver.

■ Section 3: How Does Machine Learning Work?

Machine Learning follows a three-step cycle. Understanding this cycle is the foundation of every ML project you will ever work on:

1

Collect & Teach

Gather real-world data and feed it to your model so it can learn the patterns and rules hidden within it.

2

Test the Learning

Check how well your model learned by evaluating it with new examples it has never seen before.

3

Make Predictions

Deploy your trained model to predict answers for brand new situations — this is ML in action!

■ Section 4: Meet scikit-learn

scikit-learn is a free, open-source Python library that makes building Machine Learning models simple and beginner-friendly. It is widely used by students, researchers, and professional data scientists all over the world.

■ ■ Setting Up Your Environment

Step	What to Do	Details
1	Install Python	Download from python.org — it's free and cross-platform.
2	Install scikit-learn	Open a terminal and type: <code>pip install scikit-learn</code>
3	Use Jupyter Notebook	Write and run Python code interactively. Try Google Colab for free.

■ Section 5: Our ML Project — Fruit Classifier

We will build a model that can guess whether a fruit is an **Apple** or an **Orange** based on two features: its **size** and its **colour**. This project teaches the complete ML workflow in a fun, relatable context.

■ Understanding Our Data

Feature / Label	Description	Example Values
Size (cm)	How big is the fruit? — Input feature	7, 8, 6, 9 ...
Colour (scale)	How red/orange is it? 1 = orange, 3 = red — Input feature	1, 2, 3
Fruit Type	Apple or Orange? — This is the LABEL (output we predict)	Apple, Orange

■ Decision Tree: Our Chosen Model

A **Decision Tree** makes decisions by asking a series of yes/no questions — just like a flowchart. For example: *Is the fruit big?* → Yes → *Apple!* No → *Orange!* It is easy to understand and perfect for beginners.

■ ■ Section 6: Step-by-Step Code

Step 1 — Import libraries and prepare the data

```
from sklearn.tree import DecisionTreeClassifier

# Features: [size_in_cm, colour_scale]
X = [[7, 3], [8, 3], [6, 1], [9, 1], [7, 2]]

# Labels: 0 = Apple, 1 = Orange
y = [0, 0, 1, 1, 0]
```

Step 2 — Create and train the model

```
# Create a Decision Tree classifier
model = DecisionTreeClassifier()

# Train (fit) the model on our data
model.fit(X, y)
```

Step 3 — Make a prediction

```
# Predict: what is a fruit with size=8cm and colour=3?
```

```
prediction = model.predict([[8, 3]])

if prediction[0] == 0:
    print('Prediction: Apple ■')
else:
    print('Prediction: Orange ■')
```

■ Section 7: Understanding the Results

Accuracy tells us how often the model makes the correct prediction. It is calculated as:

$$\text{Accuracy} = (\text{Number of Correct Predictions}) \div (\text{Total Predictions}) \times 100\%$$

■ Correct Predictions

90%

9 out of 10 times the model guessed correctly!

■ Incorrect Predictions

10%

Only 1 out of 10 times the model made an error.

■ Section 8: Key Vocabulary

Term	Definition
Artificial Intelligence	Technology that enables computers to mimic human thinking and problem-solving.
Machine Learning	A subset of AI where computers learn patterns from data automatically.
Model	A mathematical structure trained on data that can make predictions.
Features	The input variables (e.g., size, colour) given to the model.
Labels	The output/answer (e.g., Apple or Orange) the model is trained to predict.
Training	The process of feeding data to a model so it learns patterns.
Decision Tree	An ML algorithm that uses yes/no questions to classify data.
Accuracy	The percentage of correct predictions made by a model.
scikit-learn	A popular Python library for building ML models easily.

■ Section 9: What Did We Learn Today?

■ ML Learns from Data

Machine Learning helps computers find patterns in examples, just like you learn from practice problems at school.

■ scikit-learn is Friendly

This Python library makes building ML models simple and accessible even for complete beginners.

■ We Built a Classifier

We created a Decision Tree model that can classify fruits as Apple or Orange based on size and colour.

■ Accuracy Matters

We measured our model's performance using accuracy — the higher the accuracy, the smarter the model!

■ What's Next? Keep Exploring!

■ Try New Data

Add more fruits like bananas and change size/colour values to see how the predictions change!

■ Explore Other Models

Try K-Nearest Neighbours (KNN) or Support Vector Machines (SVM) and compare their accuracy.

■ Keep Practising

The more you code and experiment with AI, the better you'll become at building intelligent programs!

■ Thank You! Happy Learning and Coding! • Grade 6 AI & Digital Subject • Ask your teacher for help!